Data Engineering Project
Alfatraining

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1 Project Description

The LinkedIn Job Postings Dataset is a large-scale, structured collection of job advertisements published on the LinkedIn platform, capturing detailed information about labor market demand across industries, locations, and time. The dataset typically includes attributes such as job title, company name, industry, required skills, posting date, and geographic location. As a result, it provides a rich, real-world snapshot of employer hiring behavior and skill requirements at large scale.

Constructing a data warehouse from the LinkedIn Job Postings Dataset enables systematic integration, cleaning, and historical tracking of job market data to support business intelligence (BI) and strategic decision-making. A dimensional model built around a fact table of job postings, combined with dimensions such as time, location, industry, company and skills. Which allows stakeholders to analyze trends in hiring demand, emerging skills, regional labor shortages and industry growth patterns. Such data supports use cases including workforce planning, talent strategy optimization, competitive labor market analysis, and policy or investment decisions. By enabling fast, consistent, and reproducible analytics through OLAP queries and dashboards, the data warehouse transforms raw job posting data into actionable insights that can guide both operational and long-term business decisions.

1.1 Business DB- Structure and Design

The raw Linkedin popsting dataset can be found on
https://www.kaggle.com/datasets/arshkon/linkedin-job-postings/data

The database entities and relationships is presented in the following ER-diagram

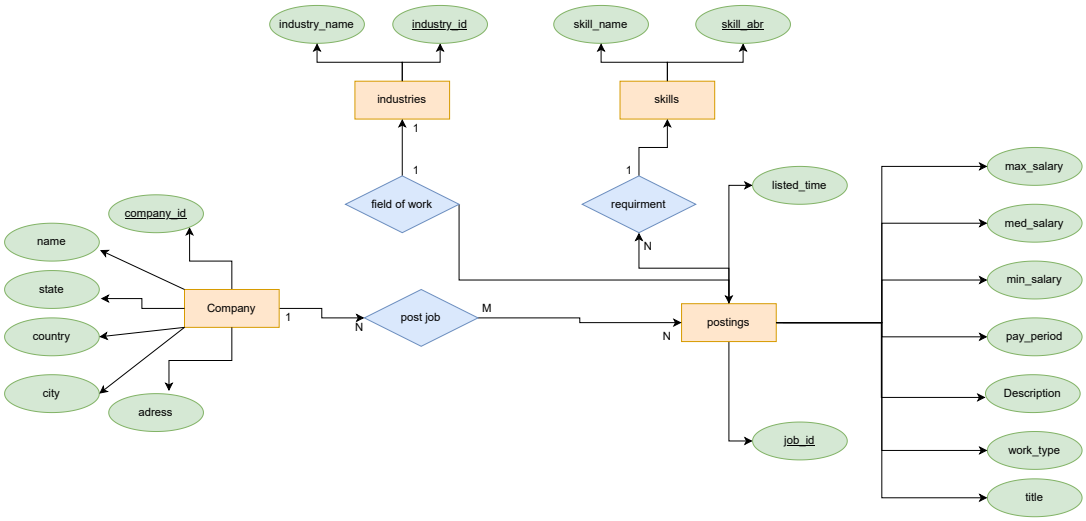

Figure 1.1: ER diagram of the Business DB

which identify the following objects :

1. **Company** : counts(24,473)

Attributes

- ► company_id(bigint,8)
  - Range(1009-103m)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (yes)
- ► name(nvarchar,100)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ► state (nvarchar,50)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)

- ► country(nvarchar,50)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ► city (Text)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ► address (Text)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)

2. **industries** : counts(422)

Attributes

- ► industries_id(smallint,4)
  - Range(1-3253)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (yes)
- ► Industry_name(nvarchar,100)
  - Range(-)
  - Repeated(0)
  - Definiteness (92%)
  - Identified (no)

3. **skills** : counts(35)

Attributes

- ► skill_abr (nvarchar,50)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (yes)
- ► skill_name (nvarchar,100)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)

4. **posting** : counts(123,849)

Attributes

- ▶ job_id (bigint,10)
  - Range(922k-3.91b)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (yes)
- ▶ company_id(bigint,8)
  - Range(1.01k-103m)
  - Repeated(0)
  - Definiteness (99%)
  - Identified (no)
- ▶ title(nvarchar,max)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ▶ listed_time (nvarchar,max)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ▶ Description (nvarchar,50)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ▶ Work type (nvarchar,50)
  - Range(-)
  - Repeated(0)
  - Definiteness (100%)
  - Identified (no)
- ▶ max_salary (nvarchar,max)
  - Range(1-120m)
  - Repeated(0)
  - Definiteness (83%)
  - Identified (no)
- ▶ med_salary (nvarchar,max)
  - Range(0-750k)
  - Repeated(0)
  - Definiteness (17%)
  - Identified (no)
- ▶ med_salary (nvarchar,max)
  - Range(1-85m)
  - Repeated(0)
  - Definiteness (83%)
  - Identified (no)

1.2 The Relational Model

A Company : $\{\underline{company.id}, name, city, state, country, address\}$
Create job post in $\Rightarrow$
Postings: $\{\underline{job.id}, listed\,tim, title, description, max\,salary, med\,salary, min\,salary, formatted\,work\,typ, \underline{company.id, industry.id, skill.abr}\}$
In the industry field $\Rightarrow$
Industries : $\{\underline{industry.id}, industry\,name\}$ Through relation field of work: $\{\underline{job.id}, \underline{industry.id}\}$
Which require $\Rightarrow$
Skills: $\{\underline{skill.abr}, skill\,name\}$ Through requirement : $\{\underline{job.id, skill.abr}\}$

1.3 Business DB Storage Requirement

companies											
	count	storage Bytes	type	length	PK	FK	Identity	Index	Unique	Not NULL	Total Bytes
company_id	24473	8	bigint		x			x	x	x	195784
name	24473	2n bytes + 2 bytes	nvarchar	100							919960
state	24473	2n bytes + 2 bytes	nvarchar	50							260516
country	24473	2n bytes + 2 bytes	nvarchar	50							96438
city	24473	16	text								387750

Industries											
	count	storage Bytes	type	length	PK	FK	Identity	Index	Unique	Not NULL	Total Bytes
industry_id	422	2	smallint		x			x	x	x	844
industry_name	422	2n bytes + 2 bytes	nvarchar	100							20024

Skills											
	count	storage Bytes	type	length	PK	FK	Identity	Index	Unique	Not NULL	Total Bytes
skill_abr	35	2n bytes + 2 bytes	nvarchar	50	x				x	x	244
skill_name	35	2n bytes + 2 bytes	nvarchar	100							20024

Figure 1.2: Business DB storage tables

postings											
	count	storage Bytes	type	length	PK	FK	Identity	Index	Unique	Not NULL	Total Bytes
job_id	122131	8	bigint		x			x	x	x	977048
company_id	122131	8	bigint			x		x	x	x	977048
title	122131	2n bytes + 2 bytes	nvarchar	max							7729212
discription	122131	2n bytes + 2 bytes	nvarchar	max							926706784
work_type	122131	2n bytes + 2 bytes	nvarchar	max							2172766
max_salary	122131	2n bytes + 2 bytes	nvarchar	max							364968
med_salary	122131	2n bytes + 2 bytes	nvarchar	max							52002
min_salary	122131	2n bytes + 2 bytes	nvarchar	max							334778
listed_time	122131	2n bytes + 2 bytes	nvarchar	max							3663930

Figure 1.3: Business DB storage tables

Total Storage (kB)				
storage per table	company	industries	skills	postings
	5,256	32	8	1,102,992
Total	1108288			

Figure 1.4: Business DB total storage

2 Data Warehouse Structure and Design

2.1 DWH Use Cases

The DWH can be used by the stakeholders to analyze trends in job market. Creating a relational data warehouse supports multiple high-value use cases. First, it enables labor market analytics, such as identifying demand trends for specific skills (e.g., Python, SQL, Machine Learning), tracking hiring intensity by geography, and analyzing industry-specific job growth over time. Second, it allows structured analysis of salary ranges by role, seniority, and region. Third, it facilitates company-level competitive analysis, including hiring speed, job category distribution, and talent acquisition strategies. Fourth, it enables time-series analysis of posting activity to detect seasonal hiring patterns.

2.2 Normalization (Up to 3rd Normal Form)

In the DWH design, one can apply normalization principles up to the Third Normal Form (3NF) in order to reduce redundancy, improve data consistency, and ensure an efficient relational structure.

First Normal Form (1NF)
To achieve 1NF, each table must contain only atomic values (no multiple values stored in one column) *i.e* each record is uniquely identifiable using a primary key. For example, instead of storing multiple cities in one field, each value is stored separately in its own row or column.

Second Normal Form (2NF)
2NF can be achieved by eliminating partial dependencies. This means that all non-key attributes depend on the entire primary key, not just part of it. This is especially important for tables with composite keys. To achieve this, data is separated into different tables where necessary. For instance, job postings company information is stored separately in company table, so that company attributes do not depend on a posting identifier. Similarly, the listed time posting attribute is normalized into Date dimension with attributes like day of week, month and year. This reduces duplication and ensures better organization. Note that some attributes from the posting table like "description" require large storage with low analysis value, thus in order to conserve disk space and computation power, such columns will be dropped.

Third Normal Form (3NF)
Finally, 3NF is implemented by removing transitive dependencies, every attribute should depend only on the primary key. For example, instead of storing city and country repeatedly in multiple posting records, a separate companies table holds location information, and the postings table references it using a foreign key.

2.3 mER Diagram and the Logical Model

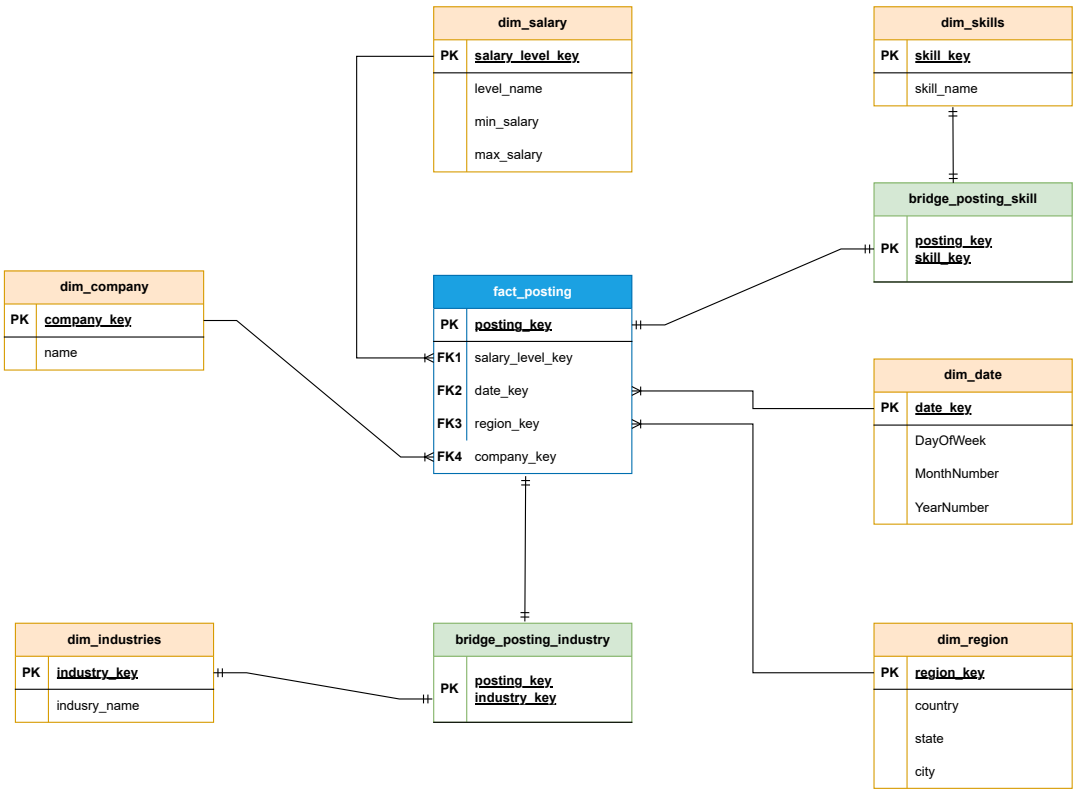

Figure 2.1: mER diagram of the proposed DWH design

2.3.1 Mapping Table and the Scripts

Please check the scripts file attached with project profile, the Python scripts used to create the DWH from the DB and ETL process are presented there. In addition, you can find the mapping table which describe the creation of the DWH from the DB tables with the suggested SCD scenarios.